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Optimality condition and stochastic subgradient for two-stage SLP

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- Properties of the recourse function
- Optimality condition
- Stochastic subgradient method



Studying the (sub) gradient of the models is useful:

1. Understanding problem structure
2. Essential in developing/understanding efficient solution algorithms
 - Stochastic gradient/approximation methods
 - Decomposition methods such as L-shaped method and extensions, Lagrange dual decomposition



- Convex hull
- Recession cone
- Normal cone
- Polyhedral function
- Support function
- Subgradient and subdifferential, basic rules
- Optimality condition



$$\begin{cases} \min & 2x_1 + 3x_2 + \mathbb{E}[Q(x, \xi)] \\ \text{s.t.} & x_1 + x_2 \leq 100 \\ & x_1, x_2 \geq 0 \end{cases}$$

$$Q(x, \xi) := \begin{cases} \min & 7y_1 + 12y_2 \\ \text{s.t.} & y_1 \geq h_1 - (2x_1 + 6x_2) \\ & y_2 \geq h_2 - (3x_1 + 3x_2) \\ & y_1, y_2 \geq 0 \end{cases}$$

- Complete fixed recourse



$$s_q(\chi) := \begin{cases} \min & 7y_1 + 12y_2 \\ \text{s.t.} & y_1 \geq h_1 - (2x_1 + 6x_2) \rightarrow \chi_1 \\ & y_2 \geq h_2 - (3x_1 + 3x_2) \rightarrow \chi_2 \\ & y_1, y_2 \geq 0 \end{cases}$$

$$\stackrel{\text{dual}}{=} \begin{cases} \max & \chi_1 \pi_1 + \chi_2 \pi_2, \\ \text{s.t.} & \pi_1 \leq 7, \\ & \pi_2 \leq 12, \\ & \pi_1, \pi_2 \geq 0. \end{cases}$$

Let $\Pi = \{\pi \in \mathbb{R}^2 : 0 \leq \pi_1 \leq 7, 0 \leq \pi_2 \leq 12\}$

- Dual feasible set: fixed, polytope
- Shall study the **dual** since its feasible set doesn't relate to x



Parameterize the RHS



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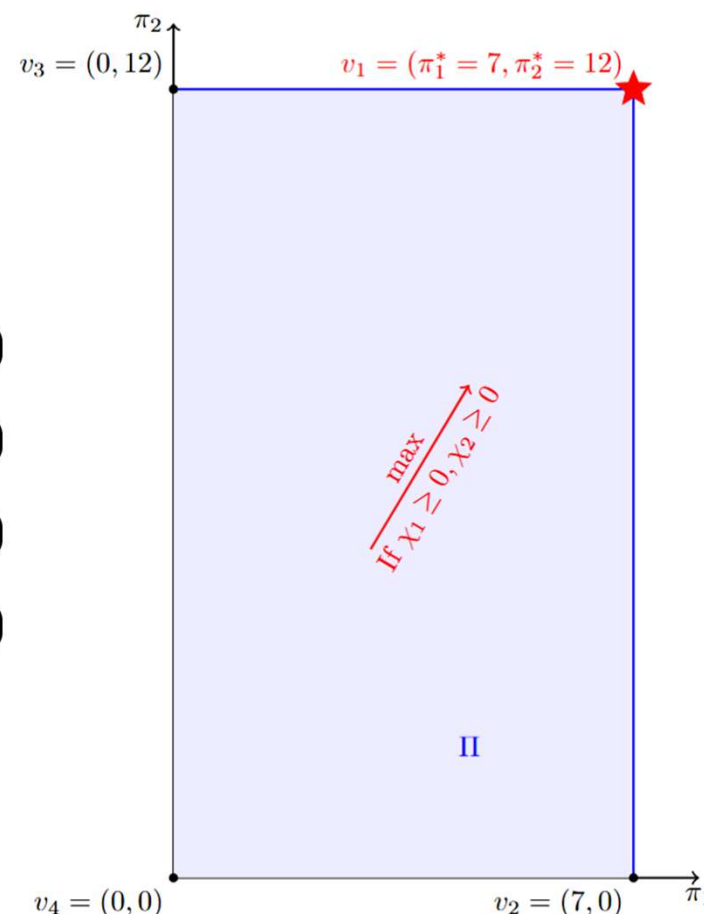
$$s_q(\chi) \stackrel{\text{dual}}{=} \begin{cases} \max & \chi_1 \pi_1 + \chi_2 \pi_2 \\ \text{s.t.} & (\pi_1, \pi_2) \in \Pi \end{cases}$$

where $\Pi = \{\pi \in \mathbb{R}^2 : 0 \leq \pi_1 \leq 7, 0 \leq \pi_2 \leq 12\}$

- Maximum of affine functions of χ and thus of x , take over a polyhedral set

$$s_q(\chi) = \begin{cases} 7\chi_1 + 12\chi_2 & \text{if } \chi_1 \geq 0, \chi_2 \geq 0 \\ 7\chi_1 & \text{if } \chi_1 \geq 0, \chi_2 \leq 0 \\ 12\chi_2 & \text{if } \chi_1 \leq 0, \chi_2 \geq 0 \\ 0 & \text{if } \chi_1 \leq 0, \chi_2 \leq 0 \end{cases}$$

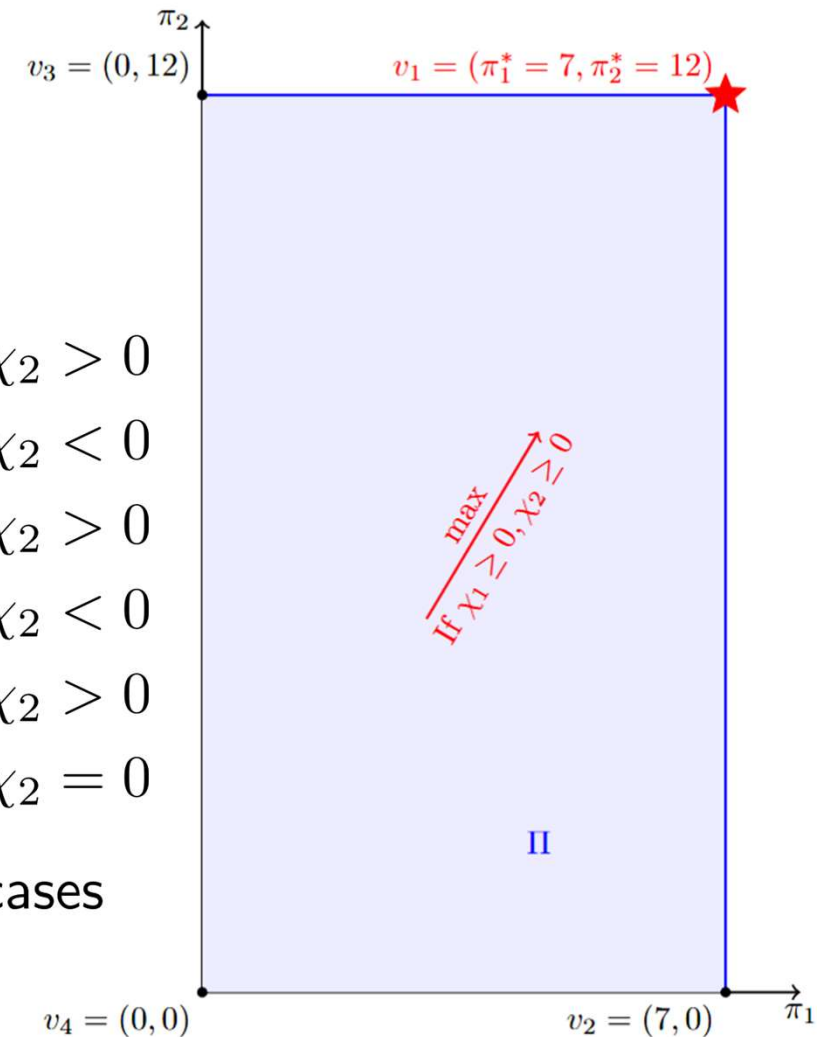
- A piecewise linear function



$$s_q(\chi) = \begin{cases} 7\chi_1 + 12\chi_2 & \text{if } \chi_1 \geq 0, \chi_2 \geq 0 \\ 7\chi_1 & \text{if } \chi_1 \geq 0, \chi_2 \leq 0 \\ 12\chi_2 & \text{if } \chi_1 \leq 0, \chi_2 \geq 0 \\ 0 & \text{if } \chi_1 \leq 0, \chi_2 \leq 0 \end{cases}$$

$$\partial s_q(\chi) = \begin{cases} \{(7, 12)\} & \text{if } \chi_1 > 0, \chi_2 > 0 \\ \{(7, 0)\} & \text{if } \chi_1 > 0, \chi_2 < 0 \\ \{(0, 12)\} & \text{if } \chi_1 < 0, \chi_2 > 0 \\ \{(0, 0)\} & \text{if } \chi_1 < 0, \chi_2 < 0 \\ \mathbf{conv}\{(7, 12), (0, 12)\} & \text{if } \chi_1 = 0, \chi_2 > 0 \\ \mathbf{conv}\{(7, 0), (7, 12)\} & \text{if } \chi_1 > 0, \chi_2 = 0 \\ \vdots & \text{boundary cases} \end{cases}$$

$$\begin{aligned} \partial s_q(\bar{\chi}) &= \mathbf{conv}\{v_l : v_l^\top \bar{\chi} = s_q(\bar{\chi})\} \\ &= \mathbf{argmax} \{ \bar{\chi}_1 \pi_1 + \bar{\chi}_2 \pi_2 : \pi \in \Pi \} \end{aligned}$$





The subdifferential of $Q(x, \xi)$



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$$Q(x, \xi) = s_q(\chi = h - Tx)$$

$$\partial Q(\bar{x}, \xi) = \frac{\partial s_q(\chi)}{\partial \chi} \cdot \frac{\partial \chi}{\partial \bar{x}}$$

$$= (-T) \cdot \operatorname{argmax} \{ \bar{\chi}_1 \pi_1 + \bar{\chi}_2 \pi_2 : \pi \in \Pi \}$$

$$= - \begin{bmatrix} 2 & 6 \\ 3 & 3 \end{bmatrix}.$$

$$\operatorname{argmax} \left\{ \begin{array}{l} (h_1 - (2\bar{x}_1 + 6\bar{x}_2))\pi_1 \\ + (h_2 - (3\bar{x}_1 + 3\bar{x}_2))\pi_2 : \pi \in \Pi \end{array} \right\}$$



- 2-stage SLP

$$\begin{cases} \min & c^\top x + \mathbb{E}[Q(x, \xi)] \\ \text{s.t.} & Ax = b, x \geq 0 \end{cases}$$

$$Q^{\text{primal}}(x, \xi) := \begin{cases} \min & q^\top y \\ \text{s.t.} & Wy = h - Tx \\ & y \geq 0 \end{cases}$$

- The dual problem of $Q(x, \xi)$

$$Q^{\text{dual}}(x, \xi) := \begin{cases} \max & \pi^\top (h - Tx) \\ \text{s.t.} & W^\top \pi \leq q \end{cases}$$

- Maximum of affine functions of x , take over a polyhedral set, thus a **polyhedral function**



$Q(x, \xi)$: composition



$$Q^{\text{primal}}(x, \xi) := \begin{cases} \min & q^\top y \\ \text{s.t.} & Wy = h - Tx \\ & y \geq 0 \end{cases}$$

$$Q^{\text{dual}}(x, \xi) := \begin{cases} \max & \pi^\top (h - Tx) \\ \text{s.t.} & W^\top \pi \leq q \end{cases}$$

- We parametrize the RHS by defining

$$\begin{aligned} s_q(\chi) &:= \inf\{q^\top y : Wy = \chi, y \geq 0\}, \quad \forall \chi \in \mathbb{R}^m \\ &= \inf\{q^\top y : \chi \in \text{pos}(W)\} \end{aligned}$$

$$\text{pos}(W) := \{\chi \in \mathbb{R}^m : Wy = \chi, \text{ for some } y \geq 0\}$$

$$\implies Q^{\text{primal}}(x, \xi) = s_q(h - Tx)$$



The function $s_q(\chi)$



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- If $\chi \in \text{pos}(W)$ the primal feasible set is nonempty.
- If the dual feasible set is also not empty, then

$$s_q(\chi) = \begin{cases} \min & q^\top y \\ \text{s.t.} & Wy = \chi \\ & y \geq 0 \end{cases} = \begin{cases} \max & \pi^\top \chi \\ \text{s.t.} & W^\top \pi \leq q \end{cases} \\ = \begin{cases} \max & \pi^\top \chi \\ \text{s.t.} & \pi \in \Pi(q) \end{cases}$$

where $\Pi(q) = \{\pi \in \mathbb{R}^m : W^\top \pi \leq q\}$ is a polyhedron

- The support function of a polyhedron, thus a **polyhedral function, hence convex!**

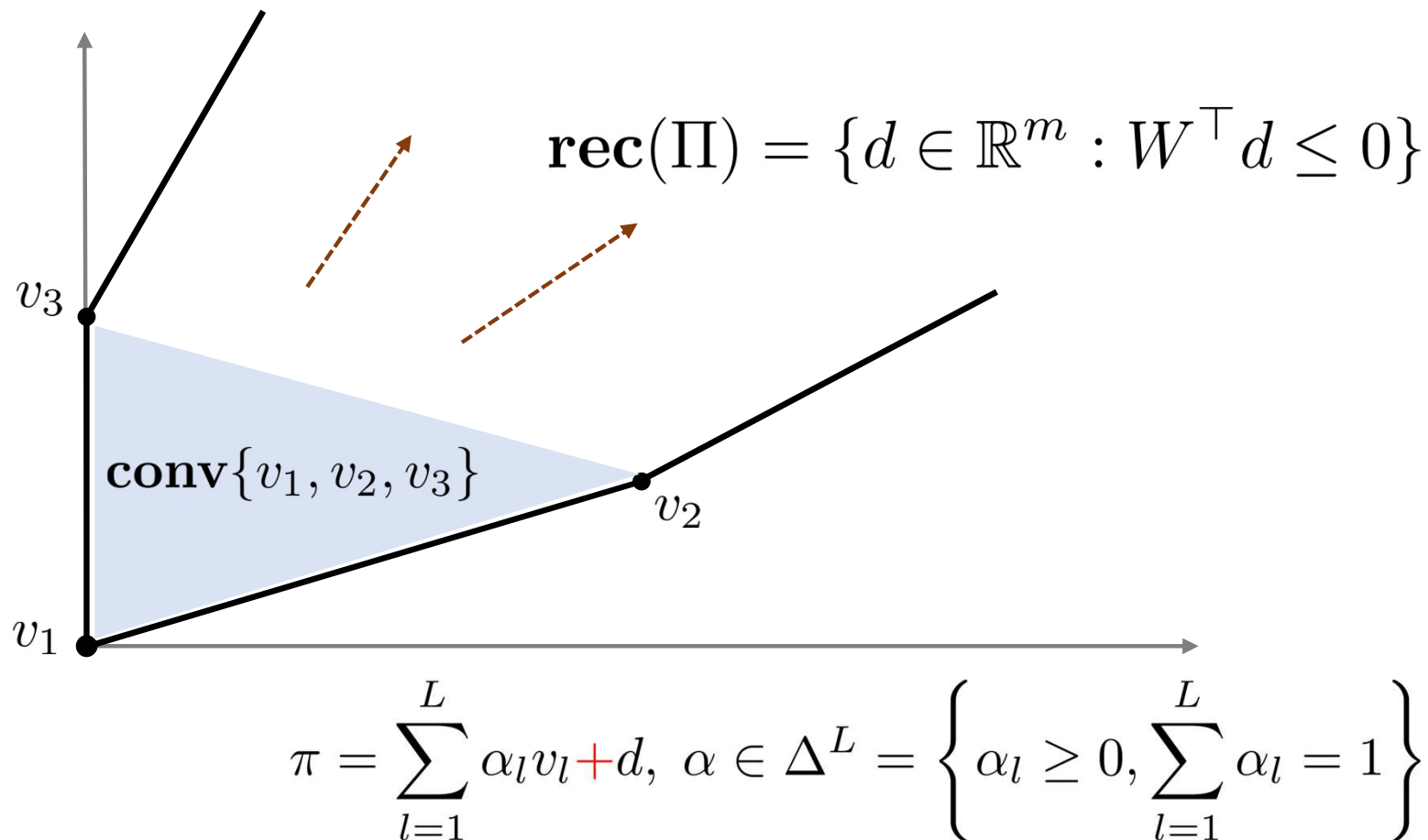


The polyhedron $\Pi(q)$



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$$\begin{aligned}\Pi(q) &= \{\pi \in \mathbb{R}^m : W^\top \pi \leq q\} \\ &= \mathbf{conv}\{v_1, v_2, \dots, v_L\} + \mathbf{rec}(\Pi)\end{aligned}$$





$$\begin{aligned} s_q(\chi) &= \begin{cases} \max & \pi^\top \chi \\ \text{s.t.} & \pi = \sum_{l=1}^L \alpha_l v_l + d \end{cases} \\ &= \begin{cases} \max & \sum_{l=1}^L \alpha_l v_l^\top \chi \\ \text{s.t.} & \alpha \in \Delta^L \end{cases} + \begin{cases} \max & d^\top \chi \\ \text{s.t.} & W^\top d \leq 0 \end{cases} \\ &= \begin{cases} \max & v_l^\top \chi \\ \text{s.t.} & l = 1, \dots, L \end{cases} + \begin{cases} 0 & d^\top \chi \leq 0 \\ +\infty & d^\top \chi > 0 \end{cases} \end{aligned}$$

= a piecewise affine function + an indicator function

$$\partial s_q(\chi_0) = \mathbf{conv}\{v_l : v_l^\top \chi_0 = s_q(\chi_0)\} = \mathbf{argmax} \{\pi^\top \chi_0 : \pi \in \Pi(q)\}$$



Proposition 1 Suppose that for given $x = x_0$ and $\xi \in \Xi$, the value $Q(x_0, \xi)$ is finite. Then $Q(x, \xi)$ is subdifferentiable at x_0 and

$$\partial Q(x_0, \xi) = -T^\top \operatorname{argmax}_{\pi \in \Pi(q)} \pi^\top (h - Tx_0)$$



of $Q(x) = \mathbb{E}[Q(x, \xi)]$

- The uncertainty parameter ξ has finite support
easier case
- The uncertainty parameter ξ has a general distribution
less easy case, depending on the recourse structure



- ξ has **finite support**: $\Xi = \{\xi_1, \xi_2, \dots, \xi_K\}$

Proposition 2 Suppose that the probability distribution of ξ has finite support $\Xi = \{\xi_1, \xi_2, \dots, \xi_K\}$ and that the expected recourse cost function $Q(x)$ has a finite value in at least one point $\bar{x} \in \mathbb{R}^n$. Then the function $Q(x)$ is polyhedral, and for any $x_0 \in \text{dom } Q = \cap_{k=1}^K \text{dom } Q(\cdot, \xi_k)$,

$$\partial Q(x_0) = \sum_{k=1}^K p_k \left\{ -T_k^\top \operatorname{argmax} \pi^\top (h_k - T_k x_0) : W_k^\top \pi \leq q_k \right\}$$



- ξ has a **general distribution**
- **If we have** (**Proposition 2.7, often satisfied in practice**):
 - i. Fixed recourse
 - ii. For a.e. q the set $\Pi(q)$ is nonempty, i.e., feasible dual
 - iii. $\mathbb{E}[||q|| \cdot ||h||] < +\infty$ and $\mathbb{E}[||q|| \cdot ||T||] < +\infty$

Then $\mathcal{Q}(x) = \mathbb{E}[Q(x, \xi)]$

- i. is well defined, and $\mathcal{Q}(x) > -\infty \forall x \in \mathbb{R}^n$
- ii. Convex, lower semicontinuous, and Lipschitz continuous on $\text{dom } \mathcal{Q}$ and
- iii. $\text{dom } \mathcal{Q} = \{x \in \mathbb{R}^n : h - Tx \in \text{pos}(W) \text{ w.p.1}\}$ is convex and closed.



- **Furthermore, if** we have $\mathcal{Q}(x)$ (**Proposition 2.8, often satisfied in practice**):
 - i. being proper, i.e., exist some x_0 such that $\mathcal{Q}(x_0) < +\infty$
 - ii. its domain has a nonempty interior.

- **Then** for any $x_0 \in \text{dom } \mathcal{Q}$

$$\partial \mathcal{Q}(x_0) = -\mathbb{E}[T^\top \pi^*(x_0, \xi)] + \mathcal{N}_{\text{dom } \mathcal{Q}}(x_0)$$

where

$$\pi^*(x_0, \xi) = \underset{\pi \in \Pi(q)}{\operatorname{argmax}} \pi^\top (h - Tx_0)$$



- ξ has finite support: $\Xi = \{\xi_1, \xi_2, \dots, \xi_K\}$

Theorem 1 Suppose that the probability distribution of ξ has finite support $\Xi = \{\xi_1, \xi_2, \dots, \xi_K\}$, and let $\bar{x}$ be a feasible solution of the two-stage SLP, i.e., $\bar{x} \in \mathcal{X}$ and $\mathcal{Q}(\bar{x})$ is finite. Then $\bar{x}$ is an optimal solution of the problem iff $\exists \pi_k^*, k = 1, \dots, K$ such that

$$\rightarrow \pi_k^* \in \operatorname{argmax} \left\{ \pi^\top (h_k - T_k \bar{x}) : W_k^\top \pi \leq q_k \right\}$$

$$\rightarrow 0 \in c - \sum_{k=1}^K p_k T_k^\top \pi_k^* + \mathcal{N}_{\mathcal{X}}(\bar{x})$$



- Use to validate a given solution, i.e., optimality certificate
- Need to solve the **2nd-stage optimal points** for every possible scenario of uncertainty
- For fixed recourse, the set/number of **2nd-stage optimal dual points** is fixed, and not depending on x



- **General case:** stochastic sampling methods
 - Sample Average Approximation (SAA) and solved using correspond deterministic opt. techniques
 - **Stochastic sub-gradient methods**



- Iterative algorithm used for minimizing a convex function, possibly **nondifferentiable**

$$f(x^*) = \min_{x \in \mathcal{X}} f(x)$$

- Use the simple iteration:

$$x^{(k+1)} = \text{Proj}_{\mathcal{X}} \left\{ x^{(k)} - \underbrace{\alpha_k}_{\text{step size}} \underbrace{g^{(k)}}_{\in \partial f(x^{(k)})} \right\}$$

- Reduce to the gradient descent for differentiable functions
- Possible that $-g^{(k)}$ is **not a descent** direction
- Keep track of the best point

$$f_{\text{best}}^{(k)} = \min\{f_{\text{best}}^{(k-1)}, f(x^{(k)})\} = \min\{f(x^{(1)}), \dots, f(x^{(k)})\}$$



- **Suppose:**
- **Lipschitz objective:** $|f(x) - f(y)| \leq L||x - y||$
- **Square summable but not summable step size**

$$\alpha_k \geq 0, \quad ||\alpha||_2^2 = \sum_{k=1}^{\infty} \alpha_k^2 < \infty, \quad \sum_{k=1}^{\infty} \alpha_k = \infty$$

- **Results:**

$$f_{\text{best}}^{(k)} - f(x^*) \leq \frac{R^2 + L^2 ||\alpha||_2^2}{2 \sum_{k=1}^{\infty} \alpha_k} \rightarrow 0 \text{ when } k \rightarrow \infty$$

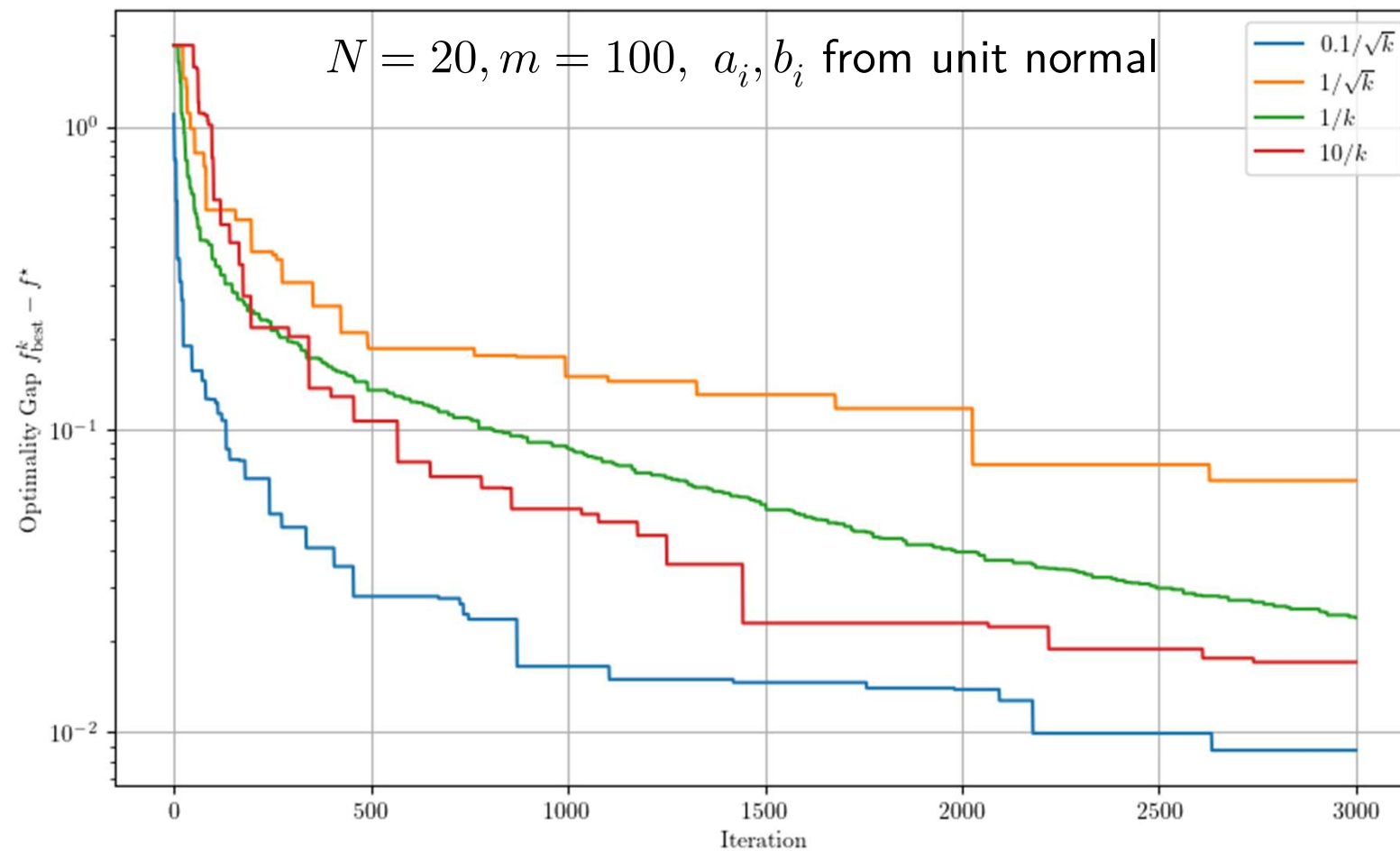
- Convergence applies also to other step size rules (see the reading material, also for the proof)



An example



$$\min_{x \in \mathbb{R}^n} f(x) = \max_{i=1, \dots, m} (a_i^\top x + b_i)$$





- For a general convex set $\mathcal{X}$, the projection operator:

$$\text{Proj}_{\mathcal{X}}(y) = \arg \min_{x \in \mathcal{X}} \|x - y\|_2^2$$

- A convex problem (QP). Sometimes more straightforward, e.g., box constraint

```
n = len(x)
y = cp.Variable(n)
objective = cp.Minimize(cp.sum_squares(y - x))
constraints = [A @ y <= b]
problem = cp.Problem(objective, constraints)
problem.solve()
return y.value
```

- **Noisy unbiased subgradient**

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. We say that a random vector $\tilde{g} \in \mathbb{R}^n$ is a **noisy (unbiased) subgradient** of f at $x \in \text{dom}(f)$ if

$$\mathbb{E}[\tilde{g}] = g \in \partial f(x),$$

i.e., for all $z \in \mathbb{R}^n$, we have

$$f(z) \geq f(x) + \mathbb{E}[\tilde{g}]^\top (z - x).$$



- In our two-stage SLP

$$\partial \mathcal{Q}(x_0) = \sum_{k=1}^K p_k \{ -T_k^\top \pi^*(x_0, \xi_k) \}$$

$$\partial \mathcal{Q}(x_0) = -\mathbb{E}[T^\top \pi^*(x_0, \xi)] + \mathcal{N}_{\text{dom } \mathcal{Q}}(x_0)$$

- Need to compute the 2nd-stage (dual) optimal points for **all the scenarios**, even the expectation, **very demanding**
- **Idea:** generate **independent** samples of scenarios (often small size), then compute a **noisy unbiased subgradient**:

$$\tilde{g}(x_0) = \frac{1}{M} \sum_{m=1}^M \{ -T_m^\top \pi^*(x_0, \xi_m) \}$$

- We can take even $M = 1$



$$x^{(k+1)} = \text{Proj}_{\mathcal{X}} \left\{ x^{(k)} - \underbrace{\alpha_k}_{\text{step size}} \underbrace{\tilde{g}^{(k)}}_{\text{noisy subgradient}} \right\}$$



- **Assume:**
- Square summable but not summable step size
- For all k , $\mathbb{E} \|\tilde{g}^{(k)}\|_2^2 \leq G^2$
- $\mathbb{E} \|x^{(1)} - x^*\|_2^2 \leq R^2$

- **Result:**

$$\lim_{k \rightarrow \infty} \mathbb{E} f_{\text{best}}^{(k)} = f(x^*)$$

$$\lim_{k \rightarrow \infty} \mathbb{P}(f_{\text{best}}^{(k)} \geq f(x^*) + \epsilon) = 0$$



$$\min_{x \in \mathbb{R}^{20}} f(x) = \mathbb{E} \left[\max_{i=1, \dots, 100} (a_i^\top x + b_i) \right]$$

$$a_i \sim \mathcal{N}(\bar{a}_i, 5I)$$

$$b_i \sim \mathcal{N}(\bar{b}_i, 5I)$$

- Using MC random sampling to compute **the noisy unbiased subgradient**

$$\rightarrow (a_{[1 \dots 100]}^m, b_{[1 \dots 100]}^m), m = 1 \dots M$$

$$\rightarrow i_m^* = \arg \max_{i=1 \dots 100} (a_i^{m \top} x + b_i^m), \forall m$$

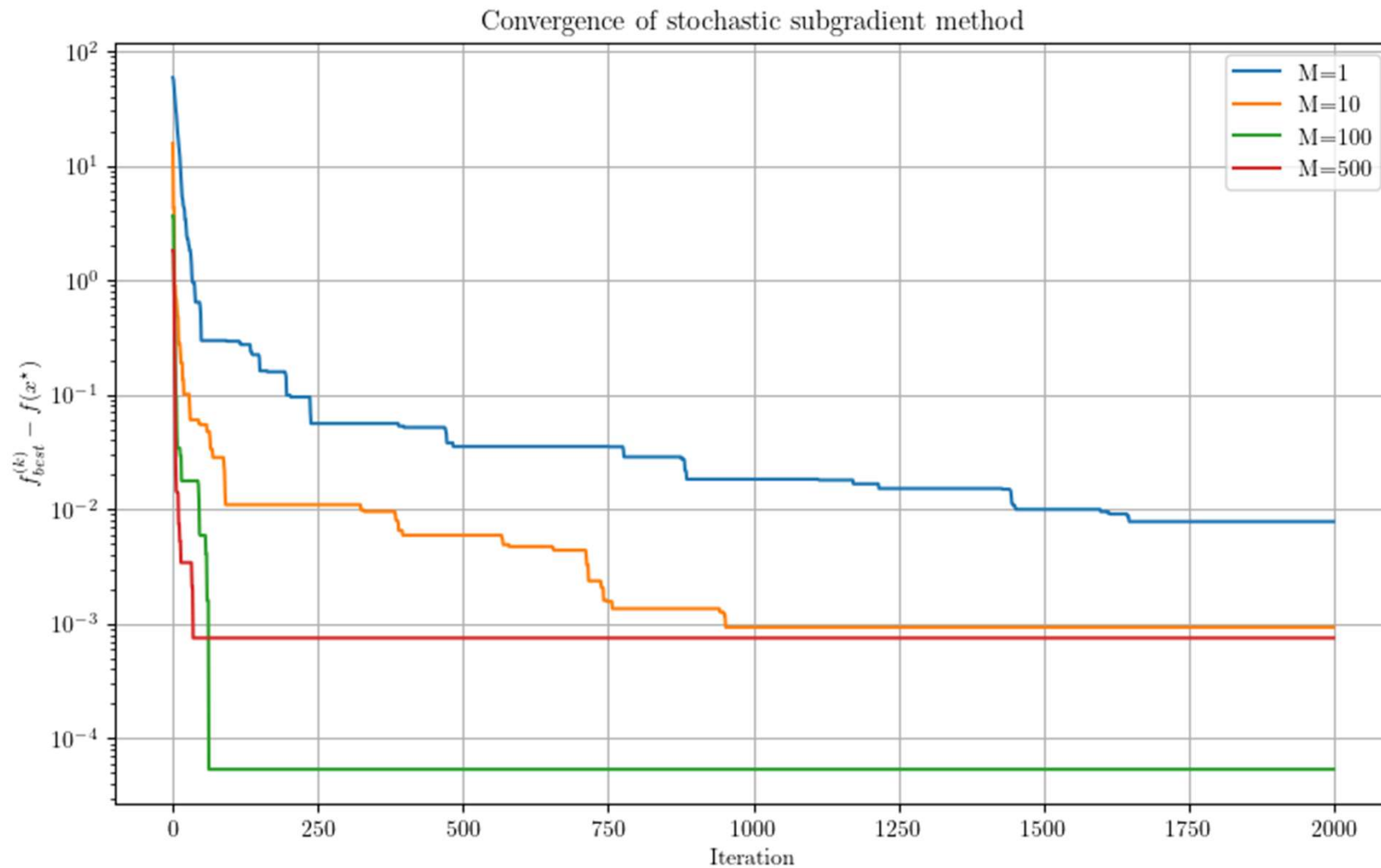
$$\rightarrow \tilde{g}(x) = \frac{1}{M} \sum_{m=1}^M a_{i_m^*}^m$$



Example



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- A lot of iterations required, the basic form often performs not so good...
- A lot of enhancements exist (and to be studied), e.g., adjust step size dynamically
- Generally slow, likely worse than decomposition methods
- not natively incorporate vanishing subgradient (when the 2nd-stage problem is infeasible)



- Computationally efficient in each iteration
 - computing subgradient and performing the update are **linear in the dimension of x** , i.e., $O(n)$, making it **scalable to high dimensions**
 - using only **a small batch of scenarios** (the SAA might be too big to be solved)
- Suitable for **online setting**: no distribution, but a new sample coming at each time
- When accuracy is not crucial



Exercise



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- Multi-location resource allocation project



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